

R1

```
R1#
R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int e0/0
R1(config-if)#ip 192.50.20.1 255.255.255.0
^
% Invalid input detected at '^' marker.

R1(config-if)#ip add 192.50.20.1 255.255.255.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#int e0/1
R1(config-if)#ip add 12.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#ip route 23.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 24.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 26.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 32.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 45.0.0.0 255.0.0.0 12.0.0.2
*Aug 25 12:57:52.020: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#ip route 45.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 172.36.0.0 255.255.0.0 12.0.0.2
R1(config)#ip route 223.0.50.0 255.255.255.0 12.0.0.2
R1(config)#exit
R1#wr
*Aug 25 12:59:25.509: %SYS-5-CONFIG_I: Configured from console by console
R1#write
Building configuration...
[OK]
R1#
```



R2

```
R2#
R2#
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int e0/0
R2(config-if)#ip add 12.0.0.2 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#int e0/1
R2(config-if)#ip add 26.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#int e0/2
R2(config-if)#ip add 23.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#int e0/3
R2(config-if)#ip add 24.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#ip route 192.50.20.0 255.255.255.0 12.0.0.1
R2(config)#ip route 223.0.50.0 255.255.255.0 26.0.0.2
R2(config)#ip route 32.0.0.0 255.0.0.0 23.0.0.2
R2(config)#ip route 45.0.0.0 255.0.0.0 24.0.0.2
R2(config)#ip route 172.36
*Aug 25 13:04:58.270: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.2
R2(config)#exit
R2#
*Aug 25 13:05:21.247: %SYS-5-CONFIG_I: Configured from console by console
R2#write
Building configuration...
[OK]
R2#
```



R3

```
R3#
R3#
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#int e0/0
R3(config-if)#
*Aug 25 13:06:09.185: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3(config-if)#ip add 23.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#int e0/1
R3(config-if)#ip add 32.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#ip route 192.50.20.0 255.255.255.0 23.0.0.1
R3(config)#ip route 12.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.2
R3(config)#ip route 24.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 26.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 45.0.0.0 255.0.0.0 23.0.0.1
R3(config)#ip route 223.0.50.0 255.255.255.0 23.0.0.1
R3(config)#exit
R3#wri
*Aug 25 13:10:21.512: %SYS-5-CONFIG_I: Configured from console by console
R3#write
Building configuration...
[OK]
R3#
```



ENG
IN

19:01
25-08-2026

R4

```
R4#
R4#
R4#
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int e0/0
R4(config-if)#ip add 24.0.0.2 255.0.0.0
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#in
*Aug 25 13:11:39.163: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R4(config)#int e0/1
R4(config-if)#ip add 45.0.0.1 255.0.0.0
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#ip route 172.36.0.0 255.255.0.0 45.0.0.2
R4(config)#ip route 192.50.20.0 255.255.255.0 24.0.0.1
R4(config)#ip route 223.0.50.0 255.255.255.0 24.0.0.1
R4(config)#ip route 12.0.0.0
% Incomplete command.

R4(config)#ip route 12.0.0.0 255.0.0.0 24.0.0.1
R4(config)#ip route 23.0.0.0 255.0.0.0 24.0.0.1
R4(config)#ip route 32.0.0.0 255.0.0.0 45.0.0.1
%Invalid next hop address (it's this router)
R4(config)#ip route 32.0.0.0 255.0.0.0 45.0.0.2
R4(config)#ip route 26.0.0.0 255.0.0.0 24.0.0.1
R4(config)#exit
R4#
*Aug 25 13:16:38.381: %SYS-5-CONFIG_I: Configured from console by console
R4#write
Building configuration...
[OK]
R4#
```



ENG
IN



19:01
25-08-2026

R5

```
R5#
R5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#int e0/0
R5(config-if)#ip add 172.36.0.1 255.255.0.0
R5(config-if)#no sh
R5(config-if)#exit
*Aug 25 13:17:56.320: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R5(config-if)#exit
R5(config)#int e0/1
R5(config-if)#ip add 32.0.0.2 255.0.0.0
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#int e0/2
R5(config-if)#ip add 45.0.0.2 255.0.0.0
R5(config-if)#no sh
R5(config-if)#exit
R5(config)# ip route 23.0.0.0 255.0.0.0 32.0.0.1
R5(config)#ip route 24.0.0.0 255.0.0.0 45.0.0.1
R5(config)#ip route 26.0.0.0 255.0.0.0 45.0.0.1
R5(config)#ip route 12.0.0.0 255.0.0.0 32.0.0.1
R5(config)#ip route 192.50.20.0 255.255.255.0 32.0.0.1
R5(config)#ip route 223.0.50.0 255.255.255.0 45.0.0.1
R5(config)#exit
R5#wr
*Aug 25 13:22:10.220: %SYS-5-CONFIG_I: Configured from console by console
R5#write
Building configuration...
[OK]
R5#
```



R6

```
R6#
R6#
R6#
R6#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#int e0/0/
^
% Invalid input detected at '^' marker.

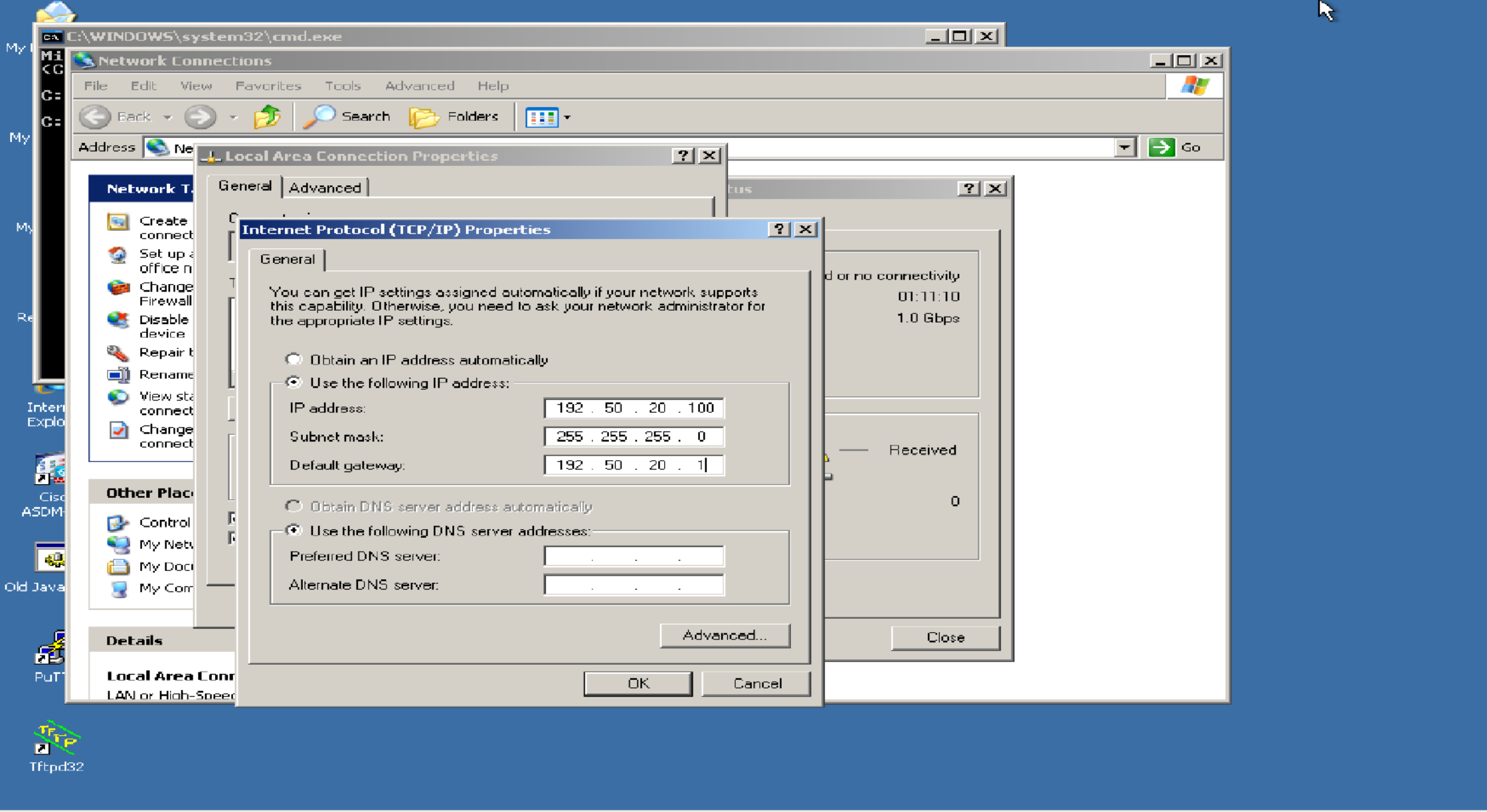
R6(config)#int e0/0
R6(config-if)#ip add 223.0.50.1 255.255.255.0
R6(config-if)#no sh
R6(config-if)#exit
R6(config)#int e0/1
R6(config-if)#ip add 26.0.0.2 255.0.0.0
R6(config-if)#no sh
R6(config-if)#exit
R6(config)#ip route 192.50.20.0 255.255.255.0 26.
*Aug 25 13:24:54.369: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R6(config)#ip route 192.50.20.0 255.255.255.0 26.0.0.1
R6(config)#ip route 172.36.0.0 255.255.0.0 26.0.0.1
R6(config)#ip route 12.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 23.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 26.0.0.0 255.0.0.0 26.0.0.1
R6(config)#no ip route 26.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 24.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 32.0.0.0 255.0.0.0 26.0.0.1
R6(config)#ip route 45.0.0.0 255.0.0.0 26.0.0.1
R6(config)#exit
R6#
*Aug 25 13:29:39.879: %SYS-5-CONFIG_I: Configured from console by console
R6#write
Building configuration...
[OK]
R6#
```



ENG
IN



19:01
25-08-2026



Microsoft Windows XP [Version 5.1.2600]

C:\WINDOWS\system32\cmd.exe

Network Connections

Local Area Connection Properties

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 223 . 0 . 50 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 223 . 0 . 50 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

Advanced...

OK Cancel

Local Area Connection Properties

General

Advanced

Speed or no connectivity 01:12:20 1.0 Gbps

Received 0

Close

Start

C:\WINDOWS\system32\cmd.exe

Network Connections

Local Area Connection Properties

Local Area Connecti...

2:39 PM

C:\WINDOWS\system32\cmd.exe

Network Connections

File Edit View Favorites Tools Advanced Help

Back Forward Search Folders

Address

Local Area Connection Properties

General Advanced

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 172 . 36 . 0 . 10

Subnet mask: 255 . 255 . 0 . 0

Default gateway: 172 . 36 . 0 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

Advanced...

OK Cancel

Local Area Connection Properties

General

01:14:33

1.0 Gbps

Received

0

Close

Start

C:\WINDOWS\system32\cmd.exe

Network Connections

Local Area Connection Properties

Local Area Connecti...

2:41 PM



C:\WINDOWS\system32\cmd.exe

```
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ncpa.cpl

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=4ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=4ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>_
```

... requires that you restart your